

Assessment 3

Assessment 3

Image Classification



Course Code: COSC2673 / COSC2793.

Weighting: This assessment is worth 40% of your final mark for this course.

Task type: Individual.

Due date and time: Wednesday, 10 June 2026, 11:59 pm AEDT (Week 14).

Feedback: Feedback will be provided via Canvas.

Click or tap on the relevant sections to learn more

▼ 1. Course Learning Outcomes Assessed

This assessment supports the following learning outcomes for COSC2673:

- CLO2: Justify the suitability of a range of machine learning methods for different kinds of problems.
- CLO3: Develop machine learning pipelines, including data preprocessing and feature engineering, for a range of applications contexts.
- CLO4: Rationalise major application areas of machine learning in contemporary professional and research contexts.
- CLO5: Articulate the ethical considerations and professional responsibilities involved in the application of machine learning.

This assessment supports the following learning outcomes for COSC2793:

- CLO2: Assess the suitability of a range of machine learning methods for different kinds of problems.
- CLO3: Design and implement machine learning pipelines, including data preprocessing and feature engineering, for a range of applications contexts.
- CLO4: Analogue major application areas of machine learning in contemporary professional and research contexts.
- CLO5: Articulate and critique the ethical considerations and professional responsibilities involved in the application of machine learning.

▼ 2. Overview of Assessment

You will work with a real-world dataset to complete a machine learning task and practice the typical machine learning workflow, which includes:

- Defining the task and understanding the prediction target.
- Reading, splitting, and pre-processing a dataset.
- Conducting **exploratory data analysis (EDA)** to understand feature distributions and potential data issues.
- Implement a number of machine learning models.
- Design and apply a consistent evaluation framework (e.g, with hold-out validation) for **model evaluation, architecture design and hyperparameter selection**.
- Support your analysis with **quantitative results and visualisations**. This should include, but not limited to:
 - performance curves (e.g., accuracy vs. hyperparameters),
 - and, for neural networks, training loss curves across epochs.
- Interpret the **behaviour of each model** based on experimental results. Your discussion should explain patterns such as underfitting, overfitting, convergence, or instability.
- **Explore** advanced techniques such as data augmentation, transfer learning, or ensembles to improve the performance of the models.
- **Evaluate** all models using consistent metrics (e.g., accuracy and F1-score) and **compare** their performance systematically.
- **Select** a final model and **justify** your choice.
- Discuss any limitations and potential real-world implications.

To complete this assignment, you will need knowledge and skills from all lectures and labs in this course. You may also use any techniques, algorithms, or models not covered in this course provided justification and a thorough explanation of these techniques, algorithms, or models if not covered in the course. Remember, developing a robust machine learning model takes time, so **start early**.

This assignment has **four** deliverables:

1. **Your Jupyter notebook**, including code, plots, results, and markdown explanations for the rationale for your approach, model design, critical analysis, and your final evaluation, along with clear instructions on how to run it. Prior to submission, please ensure all code cells execute in sequential order with the latest outputs saved. The easiest way to do this is to select "Restart & Run All" from the Kernel tab. These saved outputs will be used to verify your work.
2. **A PDF version of your Jupyter notebook**, identical to the .ipynb version.
3. **A PDF report**, detailing your machine learning workflow and rationale, i.e., introduction, definition of the task, description of the dataset, splitting, exploratory data analysis, pre-processing, machine learning techniques and model development, hyperparameter tuning, model comparison, extensions, evaluation, justification of the selected model, and in-depth analysis, ethical considerations and professional responsibilities.
4. **A 5-8-minute Recorded Video Presentation**, with microphone/camera ON, summarising your work. Your presentation should roughly, but not necessarily exactly, follow the flow of your report (see above).

You may use GenAI tools for completing your assignment. If you do use a tool, you must reference the software, declaring the details of the prompts used, including a brief description of how it was used (e.g., debugging code, clarifying concepts, improving writing). Read here about how to [Acknowledge and Reference AI](#).

You are responsible for the correctness, validity, and integrity of all submitted work. Submitting content that is largely generated by GenAI without sufficient understanding or modification is **NOT** acceptable. Plagiarism penalties apply if you did not do the work submitted. If the work you submit is highly similar to that of another student or any other source, because you both used GenAI tools or GenAI adapted from a source without proper attribution, this will also trigger plagiarism penalties, as it implies blind copying of GenAI output.

Our standard [Assessment Extensions and Late Policy](#) applies.

▼ 3. Assessment Task

The task is to develop a machine learning system that can classify histopathology images of colon cells. A basic description of histopathology can be found here (<https://www.rcpath.org/discover-pathology/news/fact-sheets/histopathology.html> ) . We will use the CRCHistoPhenotypes dataset which is a public dataset introduced in "Locality Sensitive Deep Learning for Detection and Classification of Nuclei in Routine Colon Cancer Histology Images" by Sirinukunwattana et al. in 2016, available here (<https://ieeexplore.ieee.org/document/7399414> ) , via the RMIT Library.

You will be using a modified version of the "CRCHistoPhenotypes" dataset for this task ([dataset](#) ) . The dataset in this assignment has been specifically prepared for you, and is provided on Canvas. The dataset consists of 27x27 RGB images of colon cells from 99 different patients and you are expected to use the dataset to perform two tasks:

- Classify images according to whether the given cell image represents cancerous cells or not (**isCancerous**).
- Classify images according to **cell-type**, such as: fibroblast, inflammatory, epithelial or others.

The labels of the images are in "data_labels_mainData.csv" and "data_labels_extraData.csv".

"data_labels_mainData.csv" contains data from 60 patients labelled by medical experts for isCancerous and cell-type. "data_labels_extraData.csv" contains the remaining 39 patients labelled by medical experts for isCancerous only. For the isCancerous task, you should use data from both files, "data_labels_mainData.csv" and "data_labels_extraData.csv", but for the cell-type task, you should use data from the "data_labels_mainData.csv" file.

Requirements:

- You are required to design an evaluation framework to assess model performance. This includes: (a) splitting the data into training, validation, and test; (b) selecting appropriate evaluation metrics (e.g., accuracy and F1-score). Your evaluation framework should evaluate the model generalisation to new patients. In other words, use 20% of the data as the test set, where test patients must be completely unseen (no overlap with training/validation patients).
- You should incorporate at least one unsupervised learning technique (clustering, dimensionally reduction, or both) in your exploratory data analysis (EDA) to visualise and/or understand the data. Hint: You may use the images in their natural 3-dimensional shape, flatten into a 1-dimensional shape, or otherwise.
- You must investigate at least three supervised machine learning algorithms for each of the two tasks (isCancerous and cell-type). In other words, you must develop at least three models capable of predicting

isCancerous, and at least three models capable of classifying the cell-type. You should investigate how different models perform on these classification tasks and determine which model is most appropriate. Hint: You may use the images in their natural 3-dimensional shape, flatten into a 1-dimensional shape, or otherwise.

- For at least one of your models, you should tune at least one hyperparameter of your model, such as the "learning rate" parameter in Neural Networks (set several candidate learning rates and run your model). You are required to discuss how to choose the best value for this hyperparameter, and its impact on the training and validation performance of the model.
- For at least one of your models, you should investigate at least one advanced technique to further improve the performance of your model. For example, you may use data augmentation strategies, or transfer learning to pretrain your model on the CIFAR10 dataset (resized to 27x27) and fine-tune on the CRCHistoPhenotypes dataset, or use an ensemble of models (e.g., an ensemble of decision tree, SVM, and neural network). You may use the following code if you decide to use CIFAR10 to download and transform the dataset:

```
# CIFAR
import torchvision
import torchvision.transforms as transforms

transform = transforms.Compose([
    transforms.Resize((27, 27)), # same size as CRCHistoPhenotypes
    transforms.ToTensor(),
    transforms.Normalize((0.4914, 0.4822, 0.4465), (0.2470, 0.2435, 0.2616)) # pre-computed mean and
    std
])

cifar_train = torchvision.datasets.CIFAR10(root='./cifar', train=True, download=True, transform=transform)
```

- For [COSC2793 Computational Machine Learning] only: You should investigate at least two advanced techniques to further improve the performance of your model (see previous point for a list of these techniques), explain their impact on model behaviour and performance, linking experimental results to theoretical expectations (details are in the "Assessment Submission and Feedback").
- Finally, analyse the models and results using appropriate techniques and determine how adequate your model is for these tasks in the real world. You should achieve a **macro-averaged F1-score of at least 0.90** for the isCancerous task and a **macro-averaged F1-score of at least 0.60** for the cell-type task, both for new unseen patients.
- Discuss any limitations, ethical considerations, and professional responsibilities if there are any.

The dataset herein can only be used for the purpose of this assignment. Sharing or distributing this data or using this data for any other commercial or non-commercial purposes is prohibited.

▼ 4. Timeline

To help you prepare for the assessment, please consider the following activities:

Timeline	Activities
Week 11	• Review specifications and download the dataset. • Split the dataset. • Perform exploratory data analysis (EDA), including, unsupervised learning. • Pre-process the data.
Week 12	• Develop a baseline machine learning pipeline for each task. • Evaluate baseline model performance. • Implement and compare multiple models. • Conduct hyperparameter tuning for the selected model(s).
Week 13	• Analyse and compare model performance. • Explore advanced technique(s) to improve performance. • Finalise the machine learning pipeline. • Complete the final report, video presentation, and submit!

▼ 5. Assessment Submission and Feedback

You should submit two files: (1) a single zip file containing the following and (2) a PDF of the report (i.e. the report should be submitted twice, separate and within the zip file):

1. **Your Jupyter notebook**, including code, plots, results, and markdown explanations for the rationale for your approach, critical analysis, and your final evaluation, along with clear instructions on how to run it.
2. **A PDF version of your Jupyter notebook**, identical to the .ipynb version.
3. **A PDF report**, detailing your machine learning workflow, in-depth model analysis, and a detailed written report. The report is expected to cover the following: (O) Introduction; (A) Task Definition; (B) Dataset Description and Exploratory Data Analysis (EDA); (C) Data Pre-processing; (D) Model Development and Performance Analysis (Quantitative Analysis and Visualisation); (E) Advanced Techniques (F) Model Comparison; (G) Final Model Selection, Justification, and Evaluation; (H) Ethical considerations and Professional Responsibilities (if applicable); (I) Conclusion; (J) References.
 - The report should be at **minimum 8 pages and at most 16 pages** (including all figures, tables, references, and appendices, etc).
 - For [COSC2793 Computational Machine Learning] only, the report should be at **minimum 10 pages and at most 20 pages** (including all figures, tables, references, and appendices, etc).

4. **A 5-8-minute Recorded Video Presentation**, with microphone/camera ON, summarising your work. Your presentation should roughly, but not necessarily exactly, follow the flow of your report (see above). The presentation should aim to briefly provide the following (0) Introduction; (A) Task Definition; (B) Brief Dataset Description with example EDA; (C) An Overview of the Models Developed, Justification, and Comparison; (D) Attempts to Improve Performance (Advanced Techniques); (E) Final Model Selection, Justification, and Evaluation; (G) Ethical considerations and Professional Responsibilities (if applicable); (H) Conclusion. You may submit the video directly or a link to the video file on OneDrive. If the latter, please ensure we have access to the file.

The report should address the previous list, but also the following questions:

(A) Task

- What is the task and prediction target?

(B) Dataset Description and Exploratory Data Analysis (EDA)

- What are the key characteristics of the dataset? What challenges might this dataset present?

(C) Data Pre-processing

What are the preprocessing steps? Describe how the data is prepared for modelling, including encoding categorical variables and scaling features (if applicable).

(D) Model Development and Performance Analysis (Quantitative Analysis and Visualisation)

- For each model, you must: (a) Explain briefly how it works, why it was selected (pros and cons, and why it is appropriate (or not) for this specific task/dataset); For at least one model, (b) Conduct controlled experiments by varying key hyperparameters; (c) Present quantitative results using visualisations, e.g, for Neural Networks, training loss curves across epochs; (d) Provide detailed analysis of model behaviour based on experimental results, e.g., evidence of underfitting or overfitting, sensitivity to hyperparameters, etc.

- For [COSC2793 Computational Machine Learning] only, you must link experimental results to theoretical expectations, for example, if one of your chosen models is regularised logistic regression, you must identify evidence of overfitting and through theoretical expectation, improve the regularisation of the model to circumvent the problem.

(E) Advanced Techniques

- For at least one model, you must (a) Explain why this model was selected for further tuning; (b) Explain and justify why the particular technique was chosen; (c) Analyse the technique to explain through analysis and/or visualisation how it impacts model performance.

- For [COSC2793 Computational Machine Learning] only, you must link experimental results to theoretical expectations, for example, if you choose to explore ensembling, you must explain how and why it works pointing to your experimental results.

(F) Model Comparison

- Which model performs best in the developed evaluation framework?

- Do more complex models always perform better than simpler ones in this task? Why or why not?

- For [COSC2793 Computational Machine Learning] only: Do your experimental results match what you would expect based on theory? For instance, a more complex model may fit training data better, but a simpler model may generalise better, consistent with theory.

(G) Final Model Selection, Justification, and Evaluation

- Which model is selected as the final model, and what are the test results?
 - Why this model is preferred, beyond just performance?
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▼ 6. Assessment Criteria

Component	Criteria	Weighting
Task	Assesses how clearly the machine learning task is defined, the prediction target is identified.	2%
Dataset Description and EDA	Assesses the comprehensiveness of the dataset description and quality of exploratory data analysis.	2%
Data Preprocessing	Assesses appropriateness of data handling, preprocessing, and data splitting.	6%
Model Development and In-depth Analysis	Assesses model development, experimental design, hyperparameter tuning, model evaluation, visualisations, and depth of analysis.	30%
Model Comparison, Selection, and Justification	Assesses the ability to compare models and justify the final model choice.	10%
Advances Techniques	Assesses the justification, implementation, and effectiveness of the chosen technique.	10%
Prediction Performance	Assesses the correctness and reproducibility of predictions for the test dataset, and whether the performance targets are met.	10%

Component	Criteria	Weighting
Report Quality	Assesses the clarity, organisation, and professionalism of the report, including completeness of the machine learning workflow, depth and quality of analysis, detailed discussion, and effectiveness of communication.	15%
Video Presentation	Assesses the clarity, organisation, and professionalism of the presentation, including completeness of the overview of the machine learning workflow, depth and quality of analysis, discussion, and effectiveness of communication and especially the use of the correct terminology.	15%

▼ 7. Academic Integrity

Study Support Hub: Academic skills advisors and librarians will help you with researching your topic, finding information, improving your academic writing skills, academic integrity, effective learning strategies, statistics, and more!

Academic Integrity/plagiarism: You can achieve academic integrity by honestly submitting work that is your own. Presenting work that fails to acknowledge other people's work within yours can compromise academic integrity.

Special consideration: If unexpected circumstances beyond your control have affected your ability to complete an assessment, you may be eligible to apply for special consideration.

For more information, please consult the discussion forum or contact your course coordinator.

Criteria	Ratings					Pts
Task Assesses how clearly the machine learning task is defined and the prediction target is identified.	2 to >1.6 Pts High Distinction The machine learning task and prediction target are clearly, correctly, and comprehensively defined.	1.6 to >1.2 Pts Distinction The machine learning task and prediction target are correctly defined with minor omissions.	1.2 to >0.8 Pts Credit The task and prediction target are identified but lack clarity or detail.	0.8 to >0.0 Pts Pass The task definition is incomplete or partially unclear.	0 Pts NN The task and prediction target are missing or incorrectly defined.	2 pts
Dataset Description and EDA Assesses the comprehensiveness of the dataset description and quality of exploratory data analysis.	2 to >1.6 Pts High Distinction Dataset description is comprehensive and EDA provides clear, insightful analysis with appropriate visualisations.	1.6 to >1.2 Pts Distinction Dataset description and EDA are mostly complete with useful analysis and visualisations.	1.2 to >0.8 Pts Credit Dataset description and EDA are adequate but lack depth or insight.	0.8 to >0.0 Pts Pass Dataset description and EDA are limited, superficial, or partially incorrect.	0 Pts NN Dataset description and EDA are missing or incorrect.	2 pts
Data Preprocessing Assesses appropriateness of data handling, preprocessing, and data splitting.	6 to >4.8 Pts High Distinction Data preprocessing, handling, and splitting are appropriate, well-justified, and correctly implemented.	4.8 to >3.6 Pts Distinction Data preprocessing and splitting are mostly appropriate with minor issues.	3.6 to >2.4 Pts Credit Basic preprocessing and splitting are implemented with limited justification.	2.4 to >0.0 Pts Pass Preprocessing and splitting contain significant issues or omissions.	0 Pts NN Preprocessing and splitting are missing or incorrectly implemented.	6 pts
Model Development and In-depth Analysis Assesses model development, experimental design, hyperparameter tuning, model evaluation,	30 to >24.0 Pts High Distinction Models are thoroughly developed and evaluated with strong experimental design, effective tuning, insightful	24 to >18.0 Pts Distinction Models are correctly developed and evaluated with good tuning, analysis, and visualisations.	18 to >12.0 Pts Credit Models are implemented and evaluated adequately but analysis and tuning lack depth.	12 to >0.0 Pts Pass Model development or evaluation is incomplete, weak, or inconsistently applied.	0 Pts NN Models are missing, incorrect, or lack meaningful evaluation and analysis.	30 pts

Criteria	Ratings					Pts
visualisations, and depth of analysis.	analysis, and high-quality visualisations.					
Model Comparison, Selection, and Justification Assesses the ability to compare models and justify the final model choice.	10 to >8.0 Pts High Distinction Models are comprehensively compared using appropriate metrics and the final choice is strongly justified.	8 to >6.0 Pts Distinction Models are clearly compared and the final choice is reasonably justified.	6 to >4.0 Pts Credit Basic comparison is provided with limited justification of the final model.	4 to >0.0 Pts Pass Comparison or justification is weak, unclear, or incomplete.	0 Pts NN No meaningful comparison or justification is provided.	10 pts
Advanced Techniques Assesses the justification, implementation, and effectiveness of the chosen technique.	10 to >8.0 Pts High Distinction Advanced techniques are well-justified, effectively implemented, and significantly improve outcomes.	8 to >6.0 Pts Distinction Advanced techniques are appropriately implemented and reasonably justified.	6 to >4.0 Pts Credit Advanced techniques are implemented with limited justification or effectiveness.	4 to >0.0 Pts Pass Advanced techniques are weakly implemented or poorly justified.	0 Pts NN Advanced techniques are missing, incorrect, or ineffective.	10 pts
Prediction Performance Assesses the correctness and reproducibility of predictions for the test dataset, and whether the performance targets are met.	10 to >8.0 Pts High Distinction Predictions are fully correct, reproducible, and performance targets are achieved or exceeded.	8 to >6.0 Pts Distinction Predictions are mostly correct and reproducible with strong performance.	6 to >4.0 Pts Credit Predictions are generally correct but may contain minor issues or inconsistencies.	4 to >0.0 Pts Pass Predictions contain significant issues or only partially meet requirements.	0 Pts NN Predictions are incorrect, missing, or not reproducible.	10 pts
Report Quality Assesses the clarity, organisation, professionalism, completeness of the workflow, depth of analysis, and effectiveness of communication.	15 to >12.0 Pts High Distinction The report is exceptionally clear, professional, comprehensive, and communicates analysis	12 to >9.0 Pts Distinction The report is well-organised, professional, and communicates analysis clearly.	9 to >6.0 Pts Credit The report is adequate but lacks clarity, organisation, or depth in some areas.	6 to >0.0 Pts Pass The report is difficult to follow, incomplete, or poorly communicated.	0 Pts NN The report is missing, unclear, or lacks professionalism.	15 pts

Criteria	Ratings					Pts
	effectively using appropriate terminology.					
Video Presentation Assesses the clarity, organisation, professionalism, completeness of the workflow overview, depth of analysis, discussion quality, and use of correct terminology.	15 to >12.0 Pts High Distinction The presentation is highly professional, clear, engaging, and demonstrates excellent understanding using correct terminology.	12 to >9.0 Pts Distinction The presentation is clear, organised, and demonstrates strong understanding with appropriate terminology.	9 to >6.0 Pts Credit The presentation communicates the main ideas adequately but lacks depth or polish.	6 to >0.0 Pts Pass The presentation is unclear, incomplete, or demonstrates limited understanding.	0 Pts NN The presentation is missing or ineffective.	15 pts
Total points: 100						